Epidemiology Group Project – Research Proposal
130 points total

Group Project Objectives:

- Learn about an important public health challenge (disease or health outcome),
- Develop epidemiological skills and reasoning by designing a study to examine the connection between an exposure or risk factor and this disease/health outcome, and
- Write a research proposal detailing and justifying the proposed study.

Group Project Requirements:

GROUPS

- Students work in groups of four on this project throughout the semester.
- Each group submits each assignment only once on Moodle.

DRAFT SECTIONS (35 pts total: 10 pts/section for sections #1-#3, 5 pts for section #4)

Draft sections are due earlier in the semester, followed by a final written report late in the semester. ***Note:*** Students are expected to modify these drafts as you go, so that the final written research proposal reflects what each group learned and discussed throughout the semester.

Draft Section #1 – Title; Introduction: Research Question, Background, Hypothesis
Draft Section #2 – Methods: Study Design, Population Source, Data Collection, Ethics
Draft Section #3 – Methods: Analysis, Timeline, Expected Results
Draft Section #4 – Potential Conclusions, Discussion: Strengths & Limitations

FINAL WRITTEN RESEARCH PROPOSAL (75 pts)

- Clear and concise writing, with appropriate use of graphics/tables
- 10-page maximum (plus references), single-spaced, 12-point font, 1" margins
- Note who worked on each section

FINAL PRESENTATION (20 pts)

- Presentation slides (approx. 5) should include key study design elements, biggest challenges
- 10-12-minute presentation, with additional 3-5 minutes of Q&A (15 minute total)

Outline of Epidemiology Group Project – Research Proposal

1. Title
2. Introduction
    a. Research Question:
        i. Ask a "*cause & effect*" question
        ii. Include exposure and outcome/disease
        iii. Answering question will rely on individual-level data (e.g. Do individuals who smoke get lung cancer? – NOT: Is the prevalence of smoking in different countries associated with lung cancer incidence in those countries?)
        iv. Make as focused as possible: <u>Identify a specific population/time/place</u>
    b. Background
        i. Describe disease/condition and justify why this question is important to public health. Why is it a public health problem (quantify using prevalence and/or incidence measures as appropriate)? How might the exposure affect the disease/outcome? Provide the prevalence and/or incidence of the exposure, if possible.
            1. Cite data on disease and exposure/risk factor (e.g. disease frequency [if available] and/or risk), include graphical representation of data.
            2. Summarize previous studies/literature, if possible.
    c. Hypothesis
3. Methods
    a. Study Design (e.g. RCT, cohort study, case-control study, etc.)
        i. Briefly include why this is an appropriate study design.
    b. Population Source
        i. Recruitment
            1. Inclusion/Exclusion Criteria – Who will be included/eligible to participate?
            2. How/Where will participants be recruited?
            3. How many participants do you expect in your study?
    c. Data Collection
        i. Exposure Variable
            1. What is the exposure, how will it be measured, what are the units, what is the timeframe?
        ii. Outcome Variable
            1. What is the outcome, how will it be measured, what are the units, what is the timeframe?
        iii. Potential Confounders
            1. What are the potential confounders, how will they be measured, etc?
        iv. Method of Data Collection
            1. How will data be collected? Via telephone or in-person survey? Medical record? Etc.
    d. Human Subjects/Ethical Considerations
        i. Potential Risks to Subjects
        ii. Protection Against Risks

   e. Analysis Plan
      i. What statistics are you likely to calculate based on your study and research question?
         1. Include an empty table ready to present the data you collect. (For most proposals this would be a 2x2 table.)
      ii. Describe any plans to account for potential confounders as part of the analysis.
   f. Timeline
      i. Keep this very simple. Diagram when population sample is drawn, when data collected, when analyzed.
   g. Results
      i. What type of results will you provide? (Note: you won't have actual results, since you are actually conducting this study.)
4. Discussion
   a. Strengths
      i. Including relating to methods, potential value of results to public health policies/interventions, etc.
   b. Limitations
      i. Including issues related to validity and reliability of the study results, such as ability to determine causation, potential for bias, etc.
5. *Potential* Conclusions (brief summary of what you expect to find and given this your potential next steps &/or interventions)
6. References
   a. Use AMA style to cite your sources in the text and list full source descriptions in the References section. This is **essential** in any scientific document.

Additional Information

Graphs, tables, figures – if you include:

- Gather data and create your own graphs/tables/figures. If you plan to use a graph/table/figure from someone else, please discuss with the instructor.
- Don't forget to reference and explain/interpret any graphical elements in the body of the text.
  - "In Table 2, we see…"
- Title all graphs/tables/figures. Label axes on graphs.

Citations/References:

- Use the AMA style for in text citations and your bibliography. For details, see for example: https://owl.purdue.edu/owl/research_and_citation/ama_style/index.html (note links for Print Sources and Electronic Sources for more info).

Epidemiology Final Project – Research Proposal – Paper Grading Rubric

- Clear and concise writing, with appropriate use of graphics/tables
- 10-page maximum (plus references), single-spaced, 12-point font, 1" margins
- Note who worked on each section

Criteria	Poss. Score	Your Score
Title: reflects topic		
Intro - Research Question: Clear, well-defined research question (exposure, outcome, population/time/place)		
Intro – Background: Well-supported argument for why research question is important (descriptive epi: use of statistics on prevalence and/or incidence or exposure and health outcome).		
Intro – Hypothesis: Clear and matches research question.		
TITLE + INTRODUCTION	15 pts	
Methods – Study Design: Well-supported argument for selected study design.		
Methods – Population Source: inclusion/exclusion criteria, how many/how/where participants recruited		
Methods – Data Collection: exposure var, outcome var, potential confounders, method of data collection		
Methods - Human Subjects/Ethical Concerns: potential risks, protections against risks		
METHODS – PART 1	15 pts	
Methods - Analysis Plan: measures; empty results table to show how data will be analyzed; confounders		
Methods – Timeline: graphic or table, with text as necessary.		
Methods – Results: what type of results will be provided		
METHODS – PART 2	15 pts	
Discussion – Strengths: Including relating to methods, potential value of results to public health policies/ interventions, etc.		
Discussion – Limitations: includes those affecting internal validity (chance, confounding, bias) and external validity (generalizability)		
Potential Conclusions: brief summary, potential next steps &/or interventions		
DISCUSSION	15 pts	
Proper citations of sources in text and included in **References** – American Medical Assn style.		
Overall, well-organized with clear and concise sentences; proper grammar, punctuation; proof-read; within page limit.		
REFERENCES, STYLE, MECHANICS	10 pts	
Misc. Comments		
TOTAL (excluding group participation)	**70 pts**	
Participation in the group project – average of score from all group members (1: little effort – 5: full participant)		
PARTICIPATION	5 pts	*Please see Prof. Figueroa if you want to know your project participation grade.*